

SIMATS ENGINEERING

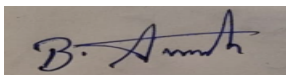
COURSE CODE: ITA1407 COURSE NAME: Ethical Hacking

Code of Conduct:

I Avinash B (192511193) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A handwritten signature in black ink on a light-colored background. The signature appears to be "B. Avinash" written in a cursive, stylized script.

CO3 AT2 port scanning challenge

Q1. Vulnerability identification and prioritization after discovering open ports

`sudo nmap -Pn -sV --script vuln --top-ports 1000 192.168.56.101`

```
(avinash@kali)-[~]
└─$ sudo nmap -Pn -sV --script vuln --top-ports 1000 192.168.56.101
[sudo] password for avinash:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:32 +0530
Stats: 0:00:39 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 24.77% done; ETC: 13:34 (0:01:25 remaining)
Stats: 0:01:37 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 31.17% done; ETC: 13:37 (0:03:12 remaining)
Stats: 0:01:42 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 31.73% done; ETC: 13:37 (0:03:18 remaining)
Stats: 0:02:40 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 38.43% done; ETC: 13:39 (0:03:59 remaining)
Stats: 0:03:47 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 46.13% done; ETC: 13:40 (0:04:13 remaining)
Stats: 0:05:28 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 53.43% done; ETC: 13:42 (0:04:37 remaining)
Stats: 0:08:06 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 58.83% done; ETC: 13:46 (0:05:33 remaining)
Stats: 0:18:02 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 77.00% done; ETC: 13:55 (0:05:20 remaining)
Stats: 0:31:54 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 97.37% done; ETC: 14:04 (0:00:00 remaining)
Nmap scan report for 192.168.56.101
Host is up (0.011s latency).
Not shown: 988 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    closed  tcpmux
8025/tcp   closed  NFS-or-IIS
8081/tcp   closed  pvuniwien
2000/tcp   closed  cisco-sccp
2004/tcp   closed  mailbox
2005/tcp   closed  deslogin
4000/tcp   closed  remoteanything
4444/tcp   closed  krb524
6000/tcp   closed  upnp
6060/tcp   open    sip?
6061/tcp   open    sip-tls?
_ftp_     _ssl-ccs-injection: No reply from server (TIMEOUT)
6667/tcp   closed  irc

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2016.66 seconds
```

Q2. Most ports appear as “filtered” — identify security mechanisms and gather more information

`sudo nmap -Pn -sS -sV --reason --top-ports 1000 192.168.56.101`

```
(avinash@kali)-[~]
└─$ TARGET="192.168.56.101"; echo "===== SYN SCAN ====="; sudo nmap -Pn -sS --top-ports 1000 "$TARGET" -oN syn.txt; echo "===== TCP CONNECT SCAN ====="; sudo nmap -Pn -sT --top-ports 1000 "$TARGET" -oN connect.txt; echo "===== SERVICE VERIFICATION ====="; sudo nmap -Pn -sV --top-ports 1000 "$TARGET"
===== SYN SCAN =====
[sudo] password for avinash:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:43 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0093s latency).
Not shown: 992 filtered tcp ports (no-response)
PORT      STATE SERVICE
1/tcp     closed tcpmux
515/tcp   closed printer
1025/tcp  closed NFS-or-IIS
1524/tcp  closed ingreslock
4444/tcp  closed krb524
5060/tcp  open  sip
5061/tcp  open  sip-tls
7001/tcp  closed afs3-callback

Nmap done: 1 IP address (1 host up) scanned in 6.51 seconds
===== TCP CONNECT SCAN =====
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:43 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0074s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
5060/tcp  open  sip
5061/tcp  open  sip-tls

Nmap done: 1 IP address (1 host up) scanned in 38.28 seconds
===== SERVICE VERIFICATION =====
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:44 +0530
Stats: 0:01:58 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 57.00% done; ETC: 13:47 (0:01:27 remaining)
Nmap scan report for 192.168.56.101
Host is up (0.0033s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
515/tcp   closed printer
5060/tcp  open  sip?
5061/tcp  open  sip-tls?

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 226.47 seconds
```

Q3. Compare TCP, UDP, and SYN port scanning techniques

`sudo nmap -Pn -sS --top-ports 100 192.168.56.101`; `sudo nmap -Pn -sT --top-ports 100 192.168.56.101`; `sudo nmap -Pn -sU --top-ports 100 192.168.56.101`

```
(avinash@kali)-[~]
└─$ sudo nmap -Pn -sS -sV --reason --top-ports 1000 192.168.56.101
[sudo] password for avinash:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:52 +0530
Stats: 0:00:26 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 12.00% done; ETC: 13:55 (0:03:03 remaining)
Nmap scan report for 192.168.56.101
Host is up, received user-set (0.0044s latency).
Not shown: 993 filtered tcp ports (no-response)
PORT      STATE SERVICE REASON VERSION
1/tcp     closed tcpmux reset ttl 126
515/tcp   closed printer reset ttl 126
2005/tcp  closed deslogin reset ttl 126
5000/tcp  closed upnp reset ttl 126
5060/tcp  open  sip? syn-ack ttl 253
5061/tcp  open  sip-tls? syn-ack ttl 253
6667/tcp  closed irc reset ttl 126

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 105.62 seconds
```

Q4. Critical services such as SSH, FTP, and database ports exposed to the Internet

`sudo nmap -Pn -sV -sC -p 21,22,3306,5432,1433,1521 192.168.56.101`

```
(avinash@kali)-[~]
$ sudo nmap -Pn -sS -sT -sU --top-ports 100 192.168.56.101
[sudo] password for avinash:
You specified more than one type of TCP scan. Please choose only one of -sA, -sB, -sT, -sF, -sI, -sM, -sN, -sS, -sW, and -sX
QUITTING!

(avinash@kali)-[~]
$ sudo nmap -Pn -sS --top-ports 100 192.168.56.101; sudo nmap -Pn -sT --top-ports 100 192.168.56.101; sudo nmap -Pn -sU --top-ports 100 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:58 +0530
Stats: 0:00:06 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 20.00% done; ETC: 13:59 (0:00:20 remaining)
Nmap scan report for 192.168.56.101
Host is up (0.0057s latency).
Not shown: 97 filtered tcp ports (no-response)
PORT      STATE SERVICE
515/tcp    closed printer
2000/tcp   closed cisco-sccp
5060/tcp   open  sip
Nmap done: 1 IP address (1 host up) scanned in 6.48 seconds
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:59 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0033s latency).
Not shown: 99 filtered tcp ports (no-response)
PORT      STATE SERVICE
5060/tcp   open  sip
Nmap done: 1 IP address (1 host up) scanned in 9.80 seconds
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 13:59 +0530
Nmap scan report for 192.168.56.101
Host is up (0.041s latency).
Not shown: 99 open/filtered udp ports (no-response)
PORT      STATE SERVICE
5353/udp   closed zeroconf
Nmap done: 1 IP address (1 host up) scanned in 16.27 seconds
```

Q5. SYN scan and TCP Connect scan produce different results — verify the findings

`sudo nmap -Pn -sS --top-ports 100 192.168.56.101; sudo nmap -Pn -sT --top-ports 100 192.168.56.101; sudo nmap -Pn -sV --top-ports 100 192.168.56.101`

```
(avinash@kali)-[~]
$ sudo nmap -Pn -sV -sC -p 21,22,3306,5432,1433,1521 192.168.56.101
[sudo] password for avinash:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-13 14:05 +0530
Stats: 0:00:07 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 88.89% done; ETC: 14:05 (0:00:00 remaining)
Nmap scan report for 192.168.56.101
Host is up.
PORT      STATE SERVICE VERSION
21/tcp    filtered ftp
22/tcp    filtered ssh
1433/tcp  filtered ms-sql-s
1521/tcp  filtered oracle
3306/tcp  filtered mysql
5432/tcp  filtered postgresql

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.74 seconds
```